Past-season, pre-season and in-season risk assessment of groin problems in male football players: a prospective full-season study

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► Additional material is published online only. To view please visit the journal online (http://dx.doi.org/10.1136/bjsports-2020-102606).

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Accepted 22 February 2021

ABSTRACT

Objective We assessed past-season, pre-season and in-season risk factors to investigate their association with an in-season groin problem in male amateur football players.

Methods Past-season groin-pain information and pre-season short-lever and long-lever adductor squeeze strength were obtained at baseline, together with anthropometrics (weight, lower limb lever length) and player age. In-season hip-related and groin-related sporting function was monitored every 4 weeks using the Sports and Recreation (Sport) subscale from the Hip And Groin Outcome Score questionnaire (HAGOS (Sport)). Groin problems, including time-loss groin injuries and groin pain irrespective of time loss, were collected over a 39-week competitive in-season. We estimated relative risk (RR), and 95% credibility interval (ICr) from logistic regressions fitted in a Bayesian framework.

Results Players (n=245) suffering from groin pain during the past-season had 2.4 times higher risk of experiencing a groin problem in the new season (2.40 RR; 95% ICr 1.5 to 3.7). This risk was reduced by 35% (0.65 RR; 95% ICr 0.42 to 0.99) per unit (N·m/kg) increase in the long-lever adductor squeeze test. Player age, short-lever squeeze test and the HAGOS (Sport) scores were not associated with the risk of a groin problem.

Conclusions Past-season groin pain increased the risk of a groin problem in the new in-season. This risk was reduced by higher pre-season long-lever adductor squeeze strength. Past-season groin-pain information and long-lever adductor squeeze strength can be quickly obtained during pre-season to identify players with an elevated risk of in-season groin problems. This may be key to reduce these problems in the new season.

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To cite: Esteve E, Casals M, Saez M, *et al*. *Br J Sports Med* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/bjsports-2020-102606

INTRODUCTION

Groin problems represent a significant health burden in male football, with a reported seasonal prevalence above 50%.[1–4] In an average week, 3–8 players in every 25-player squad will experience pain and reduced sporting function due to a groin problem.[1 2] If groin problems are only defined as time loss from football, many groin problems will not be recorded since at least one-third of players with groin problems continue to participate despite symptoms.[1 2] This potentially impacts conclusions from previous studies investigating risk factors for groin injury as the time-loss definition has been used almost exclusively.[5 6]

Previous time-loss groin injury[5–7] and reduced hip adductor strength[7 8] are both established risk factors for a new groin injury. Most previous research supports that older age is not associated with an increased risk for groin injuries.[5 6 9] These findings, however, are limited to the time-loss definition and do not account for the presence of groin pain in the previous season, irrespective of time loss. Likewise, hip adductor strength has commonly been measured using either short-lever or long-lever tests, but it is unknown whether risk estimates differ between these two tests.

Groin symptoms commonly develop gradually and fluctuate over time.[2 4 10] The Copenhagen Hip And Groin Outcome Score questionnaire (HAGOS (Sport)), a patient-reported outcome for physically active people,[11] has been proposed for predicting and monitoring hip and groin function and symptoms in football players.[9 10]

The aim of this study was to investigate whether past-season groin pain, player age, pre-season short-lever or long-lever adductor squeeze strength, and in-season hip and groin sporting function measures were associated with an in-season groin problem in male amateur football players. Second, we investigated if any of these associations differed between groin problems with and without time loss.

METHODS

Design and participants

This study used data from a cohort investigating groin and hamstring injuries in male amateur Spanish football players.[1 12 13] A convenience sample of 17 male amateur football teams (tiers IV, V and VI) were invited to participate. Players (n=363) from these teams were screened for potential risk factors during the 2015 pre-season (July–August). To be included in the study, players had to be present at baseline test and available to fully participate in the following training session. Football players who were not able to perform the test due to an injury, sickness or any physical complaint were excluded. Football players not able to understand the Spanish or English language and players under 18 years were not included. Exposure to football, groin problems and hip-related and groin-related sporting function were collected prospectively over a full competitive in-season (39

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weeks). All players were informed verbally about the purpose and procedures of the study and signed a written consent form. The reporting of this prospective cohort study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.[14]

Risk factors assessed

Past-season groin pain and age

We collected demographic (age) and anthropometric (body mass and lower-limb lever-lengths) data, in addition to retrospective information about past-season groin pain, from all included players during pre-season. A player was considered to have had past-season groin pain when they answered 'yes' to the question: Did you suffer from groin pain in the past-season?

Pre-season hip adduction squeeze strength

During pre-season, we measured players' hip adduction squeeze strength using a hand-held dynamometer (MicroFet2, Hogan Health Industries, Salt Lake City, Utah, USA) in two different testing procedures both performed by the same researcher (EE): (1) a short-lever (resistance placed between knees, feet at the examination bed and 45 degrees of hip flexion), and (2) a long-lever (resistance placed between the ankles, and 0 degrees of hip flexion). The isometric peak force (N) was obtained from one maximal repetition for both test.[15] Absolute values (N) from the respective squeeze test (short-lever and long-lever tests) from each player were normalised for body mass (kg) and lever length (m) and used as torque values (N·m/kg).

In-season hip-related and groin-related sporting function

At baseline (pre-season), and every 4 weeks in-season (overall, 11 time points), players completed the Sports and Recreation (Sport) subscale from the HAGOS questionnaire (HAGOS (Sport)).[11] Every 4 weeks scores were calculated and assigned to each player for the subsequent 4 weeks. For the first in-season week, baseline scores were assigned. In this study, we used the translated and transcultural-adapted Spanish version of the HAGOS questionnaire (available at: www.koos.nu).

Individual football exposure

Exposure to football was registered by coaching staff, using a specifically designed computer-based spreadsheet. Individual exposure to football included training exposure: defined as the number of minutes each player was involved in physical activity under the supervision of the coaching staff, and match exposure: participation in a match.[16]

Study outcomes: groin problems

A new onset of any in-season groin problem was the main outcome of interest. A groin problem was a compound outcome and included groin problems either with or without time loss, registered during the 39-week competitive period (Box 1). A new onset of any groin problem was determined using information from time-loss injury records and groin-pain surveys. A player was considered as having a new onset of a groin problem with time loss when, according to time-loss injury records, he missed at least one training session or a match due to a groin problem in a given week. A player was considered as having a new onset of a groin problem without time loss when he reported in the groin-pain survey to suffer from groin pain (yes) in a given week, but reported to be free of groin pain in the previous week and no time loss due to a groin problem was registered during the week in question.

Box 1 Registration of groin problems—with and without time loss

Groin problems with time loss

Team physiotherapists prospectively registered groin problems with time loss over the 39-week competitive in-season, as 'any complaint located in the groin leading to a player being unable to fully participate in a future training or match play' (ie, time-loss groin injury). Registration details included date of injury, and date of return to full participation. A groin problem with time loss was considered resolved when a player returned to full training participation or was available for selection for a match.

Groin problems without time loss

Groin problems without time loss were self-reported using a written groin-pain survey administered every 4 weeks. The survey asked whether players had suffered (yes/no) from groin pain in any of the past 4 weeks, with each week recorded separately. The groin-pain survey included the question: "Did you have groin pain (inguinal region including adductors, lower abdomen and pubic region) during that week?". This question was repeated for each of the last 4 weeks as part of a structured chart, including specific information regarding the dates and match opponent in each of the required weeks. An example template of the written groin-pain survey has been published previously.[1] A groin problem without time loss (ie, groin-pain episode) was considered resolved when a player reported being free of groin pain in a subsequent week.

Statistical analyses

Baseline characteristics of the included players were summarised by measures of central tendency and dispersion when the variables were quantitative, and by proportions when variables were categorical. Players were included in analyses until they were no longer part of the participating teams. Differences between players who dropped out and players who remained in the study were assessed using independent samples t-test (age, short-lever and long-lever hip adductor strength), and χ^2 (past-season groin-pain prevalence) and Mann-Whitney U (HAGOS (Sport)) test. Data normality and the homogeneity of the variance were assessed using Q-Q plots and the Kolmogorov-Smirnov test, and Levene's test, respectively. Players reporting incomplete information and/or current groin pain at baseline were not included in the analyses.

We estimated relative risk (RR) for groin problems using two multivariate models that included (1) the short-lever and (2) the long-lever squeeze test, respectively, together with past-season groin pain, age and HAGOS (Sport), all as explanatory variables. An in-season groin problem was set as the dependent variable as a mixture of its two components (ie, dependent variables): groin problems with time loss, and groin problems without time loss. Since these two components were not independent, both were modelled together in a 'two-part' model,[17 18] within a Bayesian framework using the integrated nested Laplace approximation (INLA) approach.[19] In both parts of the model, a binomial family and a logit link (equivalent to logistic regression) were assumed. Both the short-lever and the long-lever models were run again estimating separate RRs to evaluate possible differences in estimates between groin problems with and without time loss.

All models were adjusted for individual exposure to football. Team and player variables were included as random effects. All random effects were unstructured (ie, independent and identically distributed) and assumed to be normally distributed. For

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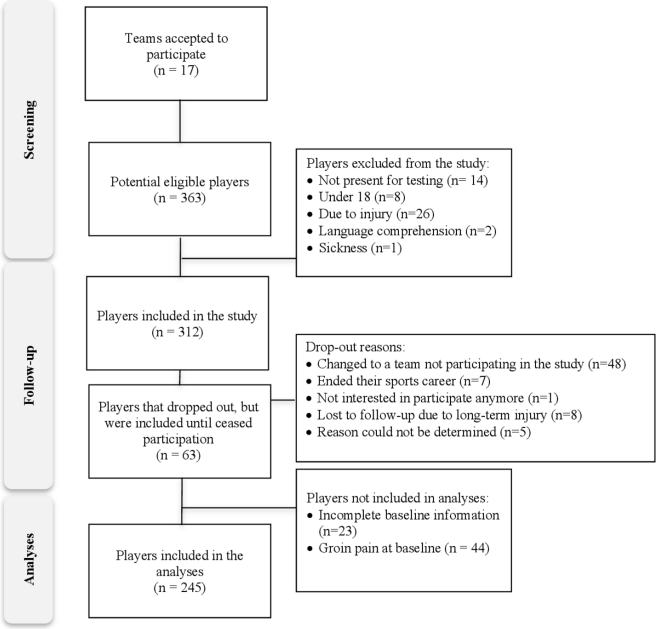

Figure 1 Flow chart of participants.

the random effects, we used priors that penalise complexity (called PC priors). These priors are robust in the sense that they do not have an impact on the results and,[19] in addition, they have an epidemiological interpretation.[20] The fixed effects (log-RR per unit increase in the covariate) were reported as posterior median of RRs, that is, exp{}, and 95% credibility interval (95% ICr). Note that in the Bayesian approach the reported 95% ICr are distributions of the sampled values, including 2.5th–97.5th percentiles.

Apart from the RRs and their 95% ICr, the probability of the parameter estimator (the log(RR)) as an absolute value being different from 0 (Prob) was also calculated (note that this is unilateral and so does not necessarily have to coincide with the credibility interval in all the cases). Unlike the p value in a usual environment, this probability allowed us to make inferences about the possible association, based on a level of significance (ie, alpha) set at 5%.

Linearity assumption was assessed by plotting outcome variables against continuous covariates. Short-lever and long-lever models were compared using the Watanabe-Akaike information criterion (WAIC).[21] Smaller values of WAIC indicated a higher performance. All analyses were performed using the free software R (V.3.4.0) (R Core Team, 2016), with the INLA package.[19]

RESULTS

Analyses included 245 players with complete baseline data (figure 1; table 1). The average (player) response rate to the groin-pain survey and HAGOS questionnaire was 93.3% (range: 89.6%–96.0%). HAGOS (Sport) scores are shown in figure 2. Two teams provided incomplete records on individual exposure. During the 39-week season, 99 players (40.4%) suffered a groin problem and 146 players (59.6%) remained free of any groin problem. The risk factor analyses included 233 in-season groin problems, of which 35 (15%) were groin problems with time loss and 198 (85%) without time loss.

Risk factors for groin problems

Estimates from multivariate models can be found in table 2. The long-lever model showed a higher performance (WAIC) compared with the short-lever model (see full models online supplemental file). Players reporting past-season groin pain had a 2.4 times higher risk of an in-season groin problem (2.40 RR; 95% ICr 1.52 to 3.54; Prob 99%). Every unit (N·m/kg) increase in pre-season long-lever squeeze adduction strength reduced the risk of a groin problem by 35% (0.65 RR; 95% ICr 0.42 to 0.99; Prob 97%). We found no association of player age or in-season HAGOS (Sport) subscale scores with risk of groin problems. Estimates from the short-lever model (table 2) showed that the short-lever squeeze test was not associated with the risk of a groin problem (0.72 RR; 95% ICr 0.39 to 1.30; Prob 86%).

Table 1 Baseline characteristics and exposure data of the 245 included players, grouped by players that remained free of groin problems and players that reported an in-season groin problem

Variables	Overall (n=245)	Not reporting groin problems (n=146)	Reporting groin problems (n=99)
Age (years)	22.9 (18–38)	22.5 (SD 3.8)	23.5 (SD 3.9)
Weight (kg)	73.7 (57–98)	74.0 (SD 7.6)	73.1 (SD 8.2)
Height (m)	1.78 (1.64–1.95)	1.78 (SD 0.6)	1.77 (SD 0.7)
Past-season groin pain*			
Yes	77 (31.4%)	37 (25.3%)	40 (40.4%)
No	168 (68.6%)	109 (74.7%)	59 (59.6%)
Short-lever squeeze strength (N·m/kg)	1.80 (SD 0.4)	1.83 (SD 0.4)	1.77 (SD 0.4)
Long-lever squeeze strength (N·m/kg)	2.79 (SD 0.5)	2.84 (SD 0.5)	2.72 (SD 0.5)
Exposure (h)	26 198	13 232	12 966
Training	21 045	10 692	10 354
Match	5152	2540	2612

Values are mean (range), unless other indicated.
*Number of players, and proportions (%) calculated in relation to the number of players in the group indicated at column top.
†

Risk factors for groin problem with and without time loss

Table 3 shows separate estimates for groin problems with and without time loss. The long-lever model showed a higher performance (WAIC) compared with the short-lever model (see full models in online supplemental file). Reporting past-season groin pain was associated with a 2.6 times higher risk of experiencing a groin problem without time loss (2.60 RR; 95% ICr 1.63 to 4.20; Prob 99%). Older age was associated with the risk of groin problems without time loss, with a 5% higher risk per year increase in age (1.05 RR; 95% ICr 1.00 to 1.11; Prob 96%). We found no association for past-season groin pain and older age with groin problems with time loss. Similar estimates were obtained for the long-lever squeeze test for groin problems either with or without time loss. We found no associations for the short-lever squeeze test, or HAGOS (Sport) with the two different groin problem definitions.

DISCUSSION

This study highlights the relevance of capturing information about the presence of groin pain in the previous season, together with pre-season long-lever hip adductor squeeze strength measures, as both are associated with the risk of a groin problem in the new in-season.

Risk factors for all groin problems

Previous studies showed that the risk of new groin injury increases with a previous time-loss groin injury,[5 6 22] and with time loss from a different injury sustained in the past-season.[22] This current study showed that past-season groin pain increased the risk of a groin problem by 2.4 times in the new season. This

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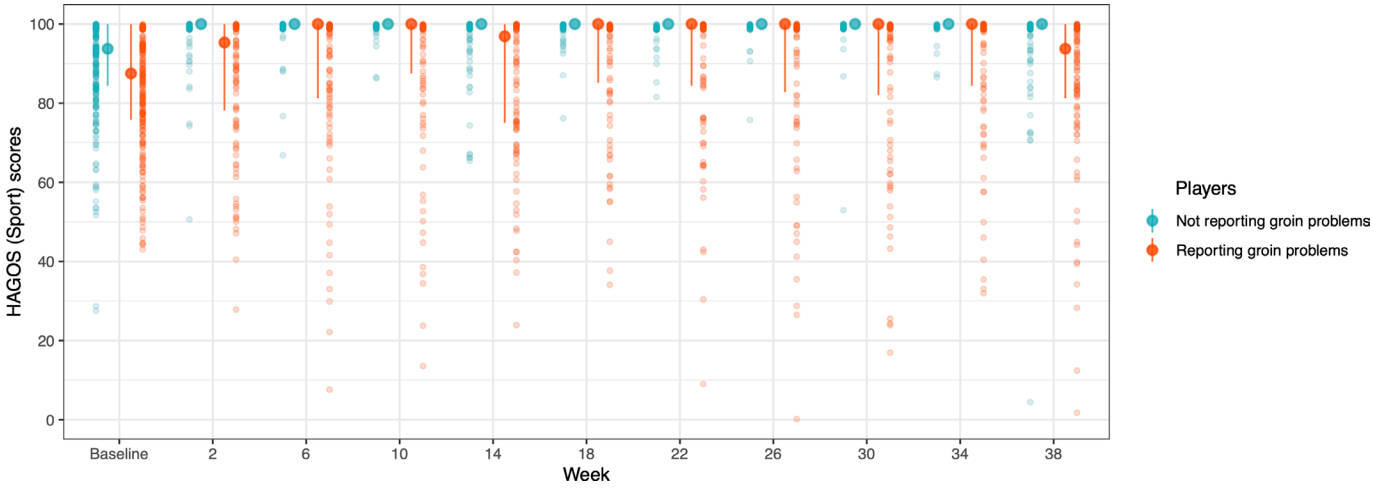

Figure 2 HAGOS (Sport) scores registered at baseline, and every 4 weeks in the in-season (39 weeks) from players reporting any groin problem during the season (at least in 1 week), and players not reporting groin problems. Scores ranging from 0 to 100, where 0 indicates extreme limitations, while 100 infers full function. Point and range values are median and quantiles (25th–75th), while dots represent raw data. HAGOS (Sport), Hip And Groin Outcome Score questionnaire.

has important implications, as the seasonal prevalence of groin pain in male football can be up to 50%.[1 3 4]

We did not identify age as a significant risk factor for a groin problem in male amateur adult football players, which is in line with previous studies.[7 8 23–25] In studies where age has been found to be a significant risk factor in initial univariate analyses, the effect disappears when adjusting for the known confounder of previous injury.[8 24]

We found that the long-lever squeeze test may play a more important role than the short-lever test on pre-season risk-factors screening. The long-lever hip adductor squeeze is more demanding on the hip adductor muscle group,[26] which are in high demand in key playing actions in football, such as kicking,[27] as well as the primary clinical entity in groin-pain cases among male football players.[25]

Baseline differences on the long-lever peak torque (table 1) between players reporting any in-season groin problem and players who remained symptom-free were minimal (4%). However, adjusted estimates (RR) revealed that the lower the peak-torque value obtained from the long-lever test the higher the risk of experiencing an in-season groin problem. By way of illustration, from a reference long-lever peak torque value of 2.8 n·m/kg (mean value from the group of players not reporting in-season groin problems), per every reduction in 0.5 N·m/kg (19% reduction), the risk of an in-season groin problem would be increased by 17.5%. This is independent of having had past-season groin pain, player age and football exposure.

Table 2 Short-lever and long-lever adductor squeeze strength multivariate models for groin problems

Variables	RR (95% ICr)	Prob.
Short-lever model		
Past-season groin pain (yes)	2.50 (1.60 to 3.40)	0.99
Age (years)	1.05 (0.99 to 1.11)	0.96
Short-lever squeeze strength (N·m/kg)	0.72 (0.39 to 1.30)	0.86
HAGOS (Sport) (0–100)	1.00 (0.99 to 1.00)	0.65
Long-lever model		
Past-season groin pain (yes)	2.40 (1.52 to 3.74)	0.99
Age (years)	1.05 (0.99 to 1.11)	0.94
Long-lever squeeze strength (N·m/kg)	0.65 (0.42 to 0.99)	0.97
HAGOS (Sport) (0–100)	1.00 (0.99 to 1.01)	0.66

HAGOS (Sport), Hip And Groin Outcome Score questionnaire; ICr, credibility interval; Prob, probability of the parameter estimator (log(RR)) being different from 0; RR, relative risk.

Table 3 Short-lever and long-lever adductor squeeze strength models with separate estimates for groin problems with and without time loss

Variables	RR (95% ICr)	Prob.
Short-lever model		
Past-season groin pain (yes)		
Time-loss groin problem	1.60 (0.75 to 3.40)	0.89
No time-loss groin problem	2.76 (1.73 to 4.44)	0.99
Age (years)		
Time-loss groin problem	1.00 (0.93 to 1.09)	0.58
No time-loss groin problem	1.06 (1.00 to 1.12)	0.98
Short-lever squeeze strength (N·m/kg)		
Time-loss groin problem	0.65 (0.26 to 1.61)	0.82
No time-loss groin problem	0.73 (0.39 to 1.37)	0.84
HAGOS (Sport) (0–100)		
Time-loss groin problem	1.00 (0.98 to 1.01)	0.79
No time-loss groin problem	1.00 (0.98 to 1.01)	0.72
Long-lever model		
Past-season groin pain (yes)		
Time-loss groin problem	1.55 (0.72 to 3.29)	0.87
No time-loss groin problem	2.60 (1.63 to 4.20)	0.99
Age (years)		
Time-loss groin problems	1.02 (0.94 to 1.10)	0.66
No time-loss groin problem	1.05 (1.00 to 1.11)	0.96
Long-lever squeeze strength (N·m/kg)		
Time-loss groin problem	0.50 (0.26 to 0.94)	0.98
No time-loss groin problems	0.67 (0.43 to 1.04)	0.96
HAGOS (Sport) (0–100)		
Time-loss groin problem	1.00 (0.99 to 1.01)	0.72
No time-loss groin problem	1.00 (0.99 to 1.00)	0.77

HAGOS (Sport), Hip And Groin Outcome Score questionnaire; ICr, credibility Interval; Prob, probability of the parameter estimator (log(RR)) being different from 0; RR, relative risk.

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Two previous studies found that lower pre-season HAGOS scores increased the risk of time-loss groin injury in the subsequent season in Gaelic[28] and professional football players.[9] Methodological differences, such as HAGOS administration frequency and the fact that these two studies were at risk of including symptomatic players at baseline,[3] could explain differences with present findings.[10]

HAGOS (Sport) was administered every 4 weeks, although the scale refers to player's hip/groin function only in the past 7 days. It is unknown if sampling more frequently would change our results, but our findings suggest that HAGOS (Sport) is more reflective of the current hip/groin function due to existing problems rather than future events.

Differences between groin problems with and without time loss

Past-season groin pain and age were associated with a higher risk of experiencing a groin problem without time loss in the new in-season, but this association was not found for groin problems with time loss. The low number of groin problems with time loss may partly explain this difference.[29] The past-season injury form and the in-season groin-pain survey referred to groin pain, irrespective of time-loss, and thus, one could expect that the past-season injury form and the in-season groin-pain survey captured groin problems of similar nature (likely due to overuse). The potential, but non-significant, increased chance of sustaining a groin problem without time loss in older athletes could be the result of greater exposure to potentially injurious loads over time and/or the recurrent and persistent nature of any previously encountered problem.

Methodological considerations and limitations

This study was conducted in a convenience sample, although there is little reason to believe that estimates would have led to different conclusions using a random sample. To reduce the impact of a potential bias through recall, the past-season injury form contained a small number of questions and limited recall to a 12-month time frame.[30 31] We found that the 69 players (28.2%) who dropped out from the study were only slightly younger (mean age: 21.7 years) compared with players who did not drop out (mean age: 23.4 years).

Individual training exposure from the two teams with incomplete exposure records was estimated by subtracting time loss (min) due to injury from team exposure, using information from time-loss injury records. We believe this had little impact on the results, as estimated training exposure (1617 hours) accounted only for 6.3% of the total individual exposure in the present study (26 198 hours).

We are cognisant that having measured hip adductor strength continuously through the season, instead of a single point at baseline, would have added additional value as this is a modifiable factor. But such multiple strength measurements were logistically, and economically not possible. However, all models were adjusted by individual exposure to football, which seems the main driver of changes in hip adductor strength.[32–34]

We were logistically not able to provide specific diagnosis for each groin-problem episode. We are aware that using a dichotomous response (groin pain yes/no) has its limitations when registering groin problems, which may fluctuate over time.[2 4 10] More comprehensive injury registration tools, such as the Oslo Sports Trauma Research Center Overuse Injury Questionnaire,[2 35] were not available in Spanish at the time the current study was performed.

What are the findings?

- The presence of groin pain in the previous season increases the risk of a groin problem in the subsequent in-season.
- Higher pre-season strength on the long-lever adductor squeeze test reduces the risk of groin problems in the new in-season.
- In-season monitoring of the hip and groin sporting function, using the Sports and Recreation (Sport) subscale from the Hip And Groin Outcome Score questionnaire (HAGOS (Sport)) every 4 weeks, was not able to detect players at risk, when adjusting for past-season groin pain and pre-season hip adductor squeeze strength.

How might it impact on clinical practice in the future?

- Risk assessments at the beginning of the season should include information on past-season groin pain irrespective of time loss.
- In pre-season, the long-lever adductor squeeze strength test is preferred over the short-lever test when assessing the risk of new in-season groin problems.
- In-season monitoring with HAGOS Sport and Recreation subscale scores registered every 4 weeks was not able to detect players with an elevated risk of a new groin problem.

Although we acknowledge the potential relevance of the severity of pain, we preferred to capture the consequences of groin-problems using HAGOS (Sport) subscale, which we consider more reflective of the functional limitations and impairments that athletes may experience. Furthermore, HAGOS (Sport) subscale showed moderate to high correlations between all subscale scores.[9] However, we are aware that using HAGOS (Sport) in the case of groin problems with time loss (ie, time-loss groin injuries) is not always ideal, as the subscale registers functional limitations in the past 7 days, and there may be days where the injury has not caused problems.

The authors acknowledge that HAGOS (Sport) scores violated the linearity assumption in relation to one of the dependent variables; groin problems without time loss. To facilitate interpretation and applicability of the present finding, we kept HAGOS (Sport) in the models in its continuous form since we found minimal changes in estimates and WAIC with or without HAGOS (Sport).

CONCLUSIONS

Players who suffered from groin pain in the previous season had an elevated risk of experiencing a groin problem in the new in-season. Higher pre-season values on the long-lever adductor squeeze test reduced the risk of experiencing groin problems in the new in-season, whereas short-lever testing, age and in-season HAGOS (Sport) were not associated with in-season groin problems. Pre-season long-lever hip adductor squeeze strength assessment and past-season groin-pain information identify players with an elevated risk of groin problems, which has implications for individualised prescription of secondary and tertiary prevention strategies, in addition to primary prevention strategies implemented at the team level.

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Acknowledgements The authors sincerely thank all players, physical therapists and members of the technical staff from teams who participated in the present study.

Contributors EE, KT, MSR, MBC and PH have contributed to the original idea of this work, design, analyses and interpretation of the data. JV-B has contributed together with EE in data collection, as well as interpretation of the data. MC and MS contributed in analyses, and visualisation and interpretation of the data. TP has contributed in interpretation of the results, writing and reviewing the manuscript. All authors have participated in writing the paper and revising it critically for important intellectual content, as well as the final approval of the version to be published.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not required.

Ethics approval This study was approved by the Comitè d'Ètica d'Investigacions Clíniques de l'Administració Esportiva de Catalunya (reference number 08/2015/CEICEGC).

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Authors are happy to share the data upon reasonable request.

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